

Setup wireless network

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A wireless network maintains devices connected to the network while allowing them to roam around freely, free of wires. In contrast, a wired network employs cables to link devices to the network. These devices are frequently desktop or laptop computers, but they can also be scanners or point-of-sale machines. Computer communication takes place on three levels:

- Hardware
- Os
- Software / application.

LAN

A local-area network is a computer network that exists at a single site, such as an office building. It can be used to connect a variety of components, such as computers, printers, and data storage devices. LANs consist of components like switches, access points, routers, firewalls, and ethernet cables to tie it all together. WI-FI is the most commonly known wireless LAN.

Setup for wireless network

One computer finds another computer in the setup process, rules, methods for communication (protocols) both computers need to agree. A client device receives the AP's beacon and translates the RF signal into digital data, which is subsequently transferred to the device for interpretation. If the user wishes to connect to the network, it can send messages to the AP attempting to join and (if security is enabled) providing the necessary credentials to indicate they are authorized to do so. These are the processes of Association and Authentication. If either of these fails, the

device will be unable to connect to the network and communicate with the AP.

Remote access

Remote enables users to connect to the systems they need when they are physically far away. This is especially important for employees who work at branch offices, are traveling or telecommute.

Router



You'll need a wireless router to set up your own Wi-Fi network. This is the gadget that will distribute the Wi-Fi signal from your Internet modem across your home. Your Internet service provider (ISP) may provide you a wireless router for a nominal monthly cost. If you've never set up a Wi-Fi network before, this may be the simplest option.

The most common way for users to connect to the Internet wirelessly is with a desktop wireless (Wi-Fi) router. Usually these routers look like small boxes with several short antennas to help broadcast signals throughout a home or an office. The farther a user is from the Wi-Fi router, the weaker the signal is. So there are usually multiple wireless routers placed throughout the workspace, in an array allowing for extended Internet coverage. A computer using IP addresses on a private network can still access the internet if router or other devices that stands between the network and the internet is using NAT (network access translation)

NAT ia a TCP/IP protocols that substitutes the public IP addresses

Configure wireless Router

Decide whether you want to use the same SSID (network name) and security settings for both the router and AP or have different ones. Using the same SSID provides a unified network experience, while different SSIDs allow clients to distinguish between the router and AP networks.

- Using your web browser, enter the **router's default IP address** into the address bar, then press **Enter**. Your router's instructions should include this information, but some of the most common addresses include **192.168.0.1**, **192.168.1.1**, and **192.168.2.1**.
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Positioning the wireless router

Place the router as close to the center of your home as possible to increase the strength of the wireless signal throughout your home.

Router WI-FI settings

Your router stores the settings for your home Wi-Fi network. To change something on your network, you have to log into your router's software, also known as firmware. From there, you can rename your network, change the password, adjust the security level, and set up or alter a variety of other options.

The process for logging into your router should be the same whether you use your internet provider's router or you purchased your own router. It should also be the same whether you use a dedicated router or a combination modem/router supplied by your provider.



Ethernet cable

Ethernet connects your computer to the internet physically, and its connections are always faster than wi-fi connections, and are usually more stable.

- Connect the ethernet cable to your router, and the other to your computer.
- An Ethernet cable, sometimes referred to as a network cable, is a cord that runs from a router, modem, or network switch to your computer, giving your device access to the local area network (LAN) — in other words, giving it internet access.
- Without walls or other objects blocking your Wi-Fi signals, you don't have to worry about sudden drops in internet speed.
- Ethernet cables come in a range of lengths and colors, but both sides of the cord are the same, regardless of the brand of cable or device you're hardwiring.

- You can utilize an Ethernet cable with a USB or USB-C adapter.

Ethernet adapter



Switch

A network switch connects devices in a network to each other, enabling them to talk by exchanging data packets.

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Switches maintain tables that match each MAC address to the port receiving the MAC address.

- Switches form the majority of network devices in modern data networks and carry huge amounts of traffic in telecommunications provider networks.
- Switches provide the wired connections to desktop computers, printers, card entry systems.

Remote access policy

The policy will enable access to the domain admins group, require certificate-based authentication, and require the highest level of encryption. Remote access Policies are described in more detail later in this chapter.

Securing your wireless network

Security is always important; with a wireless network, it's even more important because your network's signal could be broadcast outside your home. If you don't help secure

your network, people with PCs nearby could access info stored on your network PCs and use your Internet connection.

- Someone could use this info to access your router without you knowing it. To help avoid that,
 - Change the default username and password for your router
- Write down your security key and keep it in a safe place, you can also save your security key on a USB flash drive by following the instructions in the wizard.
- Use firewall